
LLM System Prompt for Task 3

You are given a part of conversation between a customer and an agent in `<conversation>` tags, where customer interacts with the agent for a specific intent. In addition, you are given the policy in `<policy>` tags that agent needs to follow in order to help the customer meet her intent.

Your task is to identify the steps that the agent has followed up to the current point from the policy and according to the next step in the policy generate an answer which is compliant with the policy.

To generate the agent's next response do the following steps:

1. read the whole conversation in `<conversation>` tag.
2. read the whole policy in `<policy>` tag.
3. based on the conversation, identify all the steps that agent has followed from the policy and get the next action from the policy to conduct.
4. generate the agent's next response, using the same language as the conversation transcript, based on the retrieved next step from the policy such that:
 - a) all the provided information should be covered in the policy's next step.
 - b) do NOT hallucinate and do NOT generate detailed information that do not exist in the provided policy's next step.
 - c) generate short responses that directly address the next best action coming from the policy.

First provide reasoning of how did you generate the next response in `<reason>` tag and then generate the next agent's response, using the same language as the conversation transcript, in `<response>` tags for the given conversation that matches with one next step in the provided policy. The generated response has to be short (less than three sentences).

Table 27: LLM system prompt for Task 3.

LLM User Prompt for Task 3

`<conversation>` *conversation* `</conversation>`

`<policy>` *policy* `</policy>`

Table 28: LLM user prompt for Task 3.

D Results

In this section, we present the results tables for experiments which were not presented in tables within the main text due to space constraints.

Model	Language	Overall	Instruction Retrieval	Begin	Middle	End
Claude3.7-Sonnet	FR	0.4041	0.8286	0.4141	0.3908	0.4065
Claude3.7-Sonnet	ES	0.4005	0.8329	0.3955	0.4100	0.3963
Claude3.7-Sonnet	EN	0.3932	0.8205	0.3959	0.3895	0.3939
Claude3.7-Sonnet	DE	0.3896	0.8195	0.3885	0.3926	0.3877
Claude3.7-Sonnet	HI	0.3882	0.8137	0.3785	0.4004	0.3867
Claude3.7-Sonnet	AR	0.3778	0.8053	0.3820	0.3734	0.3775
Claude3.7-Sonnet	ZH	0.3768	0.8096	0.3808	0.3730	0.3761
Claude3.5-Sonnet-V2	FR	0.3613	0.7372	0.3615	0.3695	0.3520
Claude3.5-Sonnet-V2	DE	0.3524	0.7276	0.3483	0.3582	0.3510
Claude3.5-Sonnet-V2	ES	0.3472	0.7227	0.3518	0.3521	0.3361
Claude3.5-Sonnet-V1	HI	0.3365	0.7260	0.3386	0.3399	0.3303
Claude3.5-Sonnet-V2	EN	0.3312	0.7311	0.3270	0.3333	0.3341
Claude3.5-Sonnet-V1	FR	0.3299	0.7112	0.3522	0.3190	0.3144
Claude3.5-Sonnet-V1	DE	0.3283	0.7088	0.3386	0.3181	0.3269
Claude3.5-Sonnet-V1	ES	0.3268	0.7073	0.3282	0.3316	0.3197
Claude3.5-Sonnet-V1	ZH	0.3216	0.7103	0.3185	0.3359	0.3095
Claude3.5-Sonnet-V2	ZH	0.3209	0.6977	0.3212	0.3129	0.3293
Claude3.5-Sonnet-V1	AR	0.3206	0.7063	0.3293	0.3146	0.3163
Claude3.5-Sonnet-V2	HI	0.3141	0.6842	0.3317	0.3102	0.2965
Qwen3-14B	EN	0.3122	0.7792	0.1645	0.2828	0.5289
Qwen3-14B	HI	0.3017	0.7582	0.1718	0.2693	0.4995
Qwen3-14B	ZH	0.2983	0.7392	0.1521	0.2654	0.5169
Qwen3-14B	FR	0.2976	0.7444	0.1540	0.2688	0.5082
Claude3.5-Sonnet-V2	AR	0.3036	0.6764	0.3007	0.3102	0.2999
Claude3.5-Sonnet-V1	EN	0.2963	0.6491	0.3019	0.2911	0.2951
Qwen3-14B	AR	0.2934	0.7135	0.1726	0.2505	0.4913
Qwen3-14B	ES	0.2874	0.7479	0.1521	0.2519	0.4952
Qwen3-14B	DE	0.2786	0.7430	0.1242	0.2558	0.4961
Gemma3-27B-IT	EN	0.2221	0.6138	0.0770	0.2144	0.4113
Gemma3-27B-IT	DE	0.2088	0.5655	0.0708	0.2044	0.3857
Gemma3-27B-IT	ES	0.1978	0.5646	0.0658	0.1834	0.3780
Gemma3-27B-IT	ZH	0.1921	0.5324	0.0550	0.1891	0.3664
Gemma3-27B-IT	HI	0.1878	0.5355	0.0701	0.1786	0.3447
Gemma3-27B-IT	FR	0.1877	0.5175	0.0441	0.1861	0.3684
Gemma3-27B-IT	AR	0.1585	0.4615	0.0433	0.1464	0.3153
Llama3.3-70B	EN	0.1358	0.2042	0.1374	0.1338	0.1360
Llama3.3-70B	ES	0.1280	0.1976	0.1327	0.1298	0.1201
Llama3.3-70B	FR	0.1146	0.1770	0.1188	0.1181	0.1056
Llama3.3-70B	DE	0.1126	0.1729	0.1262	0.1089	0.0998
Llama3.3-70B	ZH	0.0840	0.1688	0.0851	0.0780	0.0892
Llama3.3-70B	AR	0.0299	0.0653	0.0298	0.0270	0.0333
Llama3.3-70B	HI	0.0240	0.0631	0.0252	0.0227	0.0241

Table 29: Task 1 accuracy of top $k = 1$ instruction retrieval and next agent action prediction (only format f_1). Results are shown across different models and conversation languages (Arabic (AR), Chinese (ZH), English (EN), French (FR), German (DE), Hindi (HI), Spanish (ES)).

Model	Language	Overall	Instruction Retrieval	Begin	Middle	End
Claude3.7-Sonnet	EN	0.4310	0.8943	0.4481	0.4074	0.4359
Claude3.7-Sonnet	FR	0.4287	0.9022	0.4346	0.4283	0.4219
Claude3.7-Sonnet	ES	0.4246	0.9055	0.4272	0.4179	0.4286
Claude3.5-Sonnet-V2	DE	0.4237	0.8802	0.4276	0.4222	0.4204
Claude3.7-Sonnet	AR	0.4234	0.8998	0.4149	0.4314	0.4253
Claude3.7-Sonnet	DE	0.4205	0.8992	0.4327	0.4126	0.4142
Claude3.7-Sonnet	HI	0.4198	0.9003	0.4129	0.4227	0.4253
Claude3.5-Sonnet-V2	FR	0.4194	0.8838	0.4319	0.4200	0.4031
Claude3.7-Sonnet	ZH	0.4146	0.8954	0.4122	0.4070	0.4262
Claude3.5-Sonnet-V2	ES	0.4146	0.8825	0.4187	0.4109	0.4137
Claude3.5-Sonnet-V2	HI	0.4040	0.8819	0.3913	0.4131	0.4098
Claude3.5-Sonnet-V2	EN	0.3941	0.8721	0.4036	0.3904	0.3862
Claude3.5-Sonnet-V2	AR	0.3929	0.8753	0.3924	0.4048	0.3804
Claude3.5-Sonnet-V2	ZH	0.3922	0.8750	0.3901	0.3983	0.3881
Claude3.5-Sonnet-V1	FR	0.3745	0.8159	0.3808	0.3721	0.3693
Claude3.5-Sonnet-V1	DE	0.3744	0.8207	0.3750	0.3725	0.3756
Claude3.5-Sonnet-V1	HI	0.3706	0.8179	0.3769	0.3625	0.3717
Claude3.5-Sonnet-V1	ES	0.3689	0.8202	0.3731	0.3756	0.3563
Claude3.5-Sonnet-V1	AR	0.3532	0.8130	0.3595	0.3503	0.3486
Claude3.5-Sonnet-V1	ZH	0.3491	0.8066	0.3518	0.3521	0.3423
Claude3.5-Sonnet-V1	EN	0.3485	0.7830	0.3533	0.3490	0.3419
Qwen3-14B	ES	0.3340	0.8552	0.1788	0.2998	0.5651
Qwen3-14B	AR	0.3286	0.8557	0.1753	0.2824	0.5709
Qwen3-14B	EN	0.3279	0.8419	0.1668	0.2880	0.5728
Qwen3-14B	DE	0.3273	0.8441	0.1548	0.3041	0.5680
Qwen3-14B	HI	0.3207	0.8376	0.1560	0.3015	0.5473
Qwen3-14B	ZH	0.3151	0.8260	0.1513	0.2932	0.5434
Qwen3-14B	FR	0.3122	0.8199	0.1389	0.2906	0.5521
Gemma3-27B-IT	HI	0.2540	0.7230	0.0786	0.2558	0.4706
Gemma3-27B-IT	EN	0.2528	0.7283	0.0809	0.2414	0.4797
Llama3.3-70B	EN	0.2497	0.6612	0.2620	0.2366	0.2488
Gemma3-27B-IT	FR	0.2444	0.6970	0.0592	0.2227	0.4990
Gemma3-27B-IT	DE	0.2408	0.6961	0.0728	0.2248	0.4677
Gemma3-27B-IT	AR	0.2364	0.7102	0.0813	0.2209	0.4470
Llama3.3-70B	FR	0.2353	0.6872	0.2512	0.2248	0.2271
Gemma3-27B-IT	ES	0.2349	0.6885	0.0677	0.2044	0.4769
Llama3.3-70B	DE	0.2307	0.6578	0.2194	0.2423	0.2319
Llama3.3-70B	ES	0.2305	0.6859	0.2357	0.2322	0.2223
Gemma3-27B-IT	ZH	0.2262	0.6837	0.0565	0.2266	0.4373
Llama3.3-70B	ZH	0.1293	0.4207	0.1277	0.1216	0.1398
Llama3.3-70B	AR	0.0511	0.1906	0.0534	0.0449	0.0550
Llama3.3-70B	HI	0.0390	0.1769	0.0430	0.0375	0.0357

Table 30: Task 1 accuracy of top $k = 2$ instructions retrieval and next agent action prediction (only format f_1). Results are shown across different models and conversation languages (Arabic (AR), Chinese (ZH), English (EN), French (FR), German (DE), Hindi (HI), Spanish (ES)).